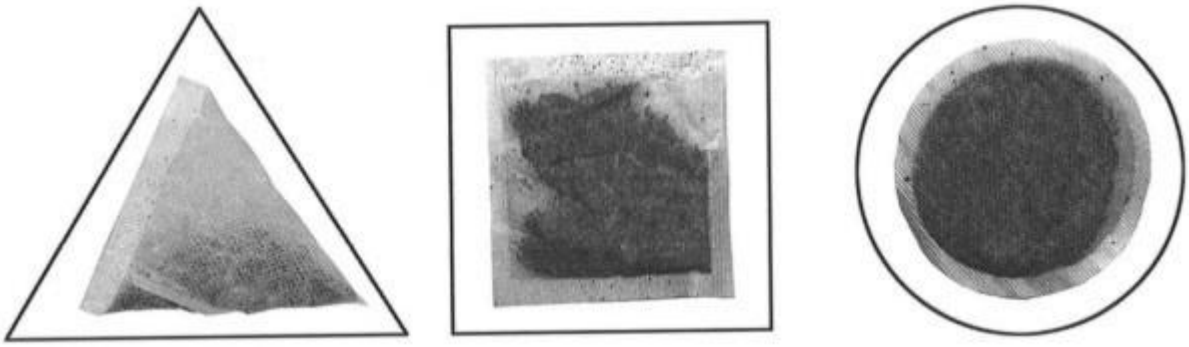
Q1. Tea bags are made in different shapes.



triangle **square** **circle**

Some pupils want to find out which shape of tea bag lets tea dissolve most quickly. They make two plans for their investigation as shown below.

FIRST PLAN
We will use 3 tea bags and 3 beakers

SECOND PLAN
Collect three beakers.
Collect three different tea bags.
Put one tea bag in each beaker.
Add 150 cm^3 of water at 65°C.
Keep the temperature of the water the same.
Measure the time taken for the tea to dissolve.
Find out which is the quickest for making tea.

(a) How is the second plan better than the first plan?

.....

.....

1 mark

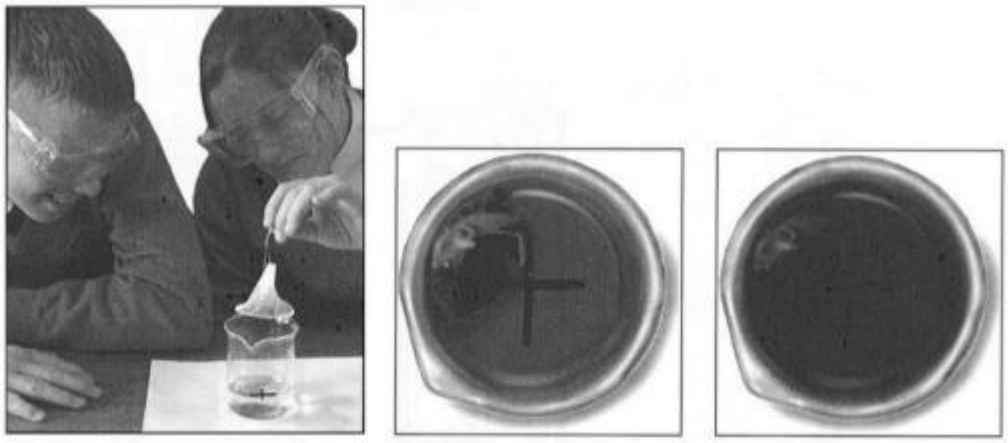
(b) Why should they take care when they add hot water at 65°C to the tea bags?

.....

.....

1 mark

(c) Ben and Vicky drew a cross on some paper. They put each beaker, in turn, over the cross. They poured hot water into the beaker, dropped in the tea bag and watched the water change colour.



To see which shape of tea bag let the tea dissolve the quickest, they measured the time until the liquid was too dark for them to see the cross.

How did the cross help to make their test more accurate?

.....

1 mark

(d) (i) They recorded their measurements in a table as shown below.

shape of tea bag	**time taken until cross cannot be seen (minutes)**
triangle	8
square	15
circle	10

Which part of their investigation was recorded in the table?
Tick the correct box.

explanations ☐ results ☐

conclusions ☐ plans ☐

1 mark

(ii) Give the **three** shapes of tea bags in the order in which the tea dissolved.
Use the table above to help you.

quickest _____ _____ _____ slowest

1 mark

maximum 5 marks

Mark schemes

Q1.

(a) any **one** from

- it describes how they will carry out their investigation
- it has more information **or** detail
- it includes a fair test
- it includes measurement

accept a description which identifies a factor to be kept constant

*accept 'the second plan includes apparatus to be used **or** a measurement **or** a comparison'*

accept the converse of any marking point

accept a statement referring to any of the points in the second plan

accept answers which describe a consequence of the test not being fair

1 (L4)

(b) to avoid scalding **or** burning themselves

accept 'it is very hot'

accept 'to avoid spilling'

credit may be given for answers which, although not accurate, imply that the water is at a high temperature eg 'it is nearly boiling'

1 (L4)

(c) any **one** from

- it allowed them to compare the times for different tea bags
- it told them when the measurement was completed
- so they knew when to stop

accept 'as soon as it has gone they stopped timing'

accept 'so they know how long it takes'

accept 'the cross let them see when the tea produced by the 3 bags was the same'

accept 'so they could stop at the right time'

accept 'it tells them when they have dissolved the same'

*answers must indicate that the cross shows when the teas are the same colour **or** allows a measurement to be made*

'it made it fair' is insufficient

1 (L3)

(d) (i) results ✓

if more than one box is ticked, award no mark

1 (L4)

(ii) triangle circle square

accept a drawing of a triangle, a circle and a square

***all three** answers are required in the correct order*

1 (L3)

[5]